

Degree B-Tech, Semester 3rd (Backlog)
SUMMER SEMESTER (MID-SEMESTER) EXAMINATION, June, 2024

Course Code- EIECC08/ECECC08/INECC01

Course Title- Digital Circuit and System

Time-1:30 Hours

Max. Marks- 15

Note: - Attempt all questions in given order only. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Perform the following conversion : 1) (1010111) gray to binary 2) (01101011) excess 3 to decimal 3) (237.8) octal to decimal	1.5	CO1
1b	Perform the following addition using 2's compliment method (+29) + (+19)	1.5	CO1
2a	Realise a JK flip flop using SR flip flop.	1.5	CO3
2b	Solve the following using rules of boolean algebra $Y = A'B + ABD + AB'CD' + BC$	1.5	CO2
3a	Implement a 16X1 multiplexer using 4X1 multiplexer.	1.5	CO2
3b	Find the minimum SOP for the given function using K maps $Y = \sum m(0, 2, 3, 6, 7) + \sum d(8, 10, 11, 15)$	1.5	CO2
4a	Implement a full subtractor circuit using a PROM	1.5	CO2
4b	Write a short note on error detecting and correcting codes	1.5	CO2
5a	Design a mod 10 counter using D flip flops.	1.5	CO3
5b	Design and explain 4 bit Serial in Parallel Out (SIPO) synchronous counter.	1.5	CO3